difference of 13 basis points represents an even more significant fraction of the 23 basis point average half spread.

Table 5. Size of Change in NBBO

Round Lot Tier[1,2]	Best Bid: Average Price Change ($) [3]	Best Ask: Average Price Change ($)[3]	Average Difference in Quoted Half Spread (%)[4]	SIP: Average Quoted Percent Half Spread (%)
1. $0-$250 (100 shares)	n/a	n/a	n/a	n/a
2. $250.01 - $1,000 (40 shares)	0.64	0.89	0.07	0.24
3. $1000.01 - $10,000 (10 shares)	2.48	2.81	0.13	0.23
4. $10,000.01+ (1 share)	n/a	n/a	n/a	n/a

1. Tier based on the stock's prior month's average closing price in April 2020.

2. Twelve stocks trade in round lots different than 100 shares and are excluded.

3. Conditional on a the instance of a smaller quote, stock-day average price improvement is calculated using MIDAS data, which consists of hourly snapshots from 10:30am to 3:30pm for each trading day in May 2020. Calculation is based on the difference between the best bid / best ask calculated under the new round lot tier definition (source: direct feeds) compared to the NBBO based on the current 100 share round lot criteria (source: SIP)

4. Conditional on a the instance of a smaller quote (bid or ask), stock-day average difference in percent quoted half spread is calculated by SIP NBBO quoted percent half spread minus the new percent quoted half spread under the proposed round lot tier criteria. Quoted half spread is defined by: Quoted half-spread = $QS_{it} = 100 * (Ask_{it} – Bid_{it}) / (2*M_{it})$, where M is the midpoint between the best bid and best ask.

In the Proposing Release, the Commission qualitatively discussed that the change in round lot size could cause the NBBO and protected quotes to widen for the twelve stocks that currently have a round lot size less than 100 shares.[1837] However, one commenter stated that the Commission did not analyze the effects of the change in round lot size and protected quotes on

[1837] See Proposing Release, 85 FR at 16824, 16830-1.

these twelve stocks.[1838] In response to this comment, the Commission quantitatively analyzed the effects of the revised definition of round lot size on these stocks using data from one week of trading in October 2020 and confirmed that the NBBO would widen in some of these stocks.[1839] The analysis showed that the round lot size will not change for four of these stocks, so their NBBO will not change. However, for eight of these stocks the round lot size would increase. In these 8 stocks, the analysis showed that, on average, the NBBO would widen 97.1% of the time under the new round lot definition. In the instances in which the NBBO was wider, the Commission found that the NBBO half-spread increased by an average of 3.66% in these stocks.

(ii) Effects on Market Participants

For stocks priced above $250, the new round lot definition will result in the inclusion of quotes at better prices in core data that were previously excluded from being reported because they consisted of too few shares. This will make these quotes visible to anyone who subscribes to core data, thereby improving transparency.[1840] The Commission believes that this will create an economic benefit for market participants who currently rely exclusively on SIP data to obtain

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1838 See NYSE Letter II at 6.

1839 The round lot tiers for these twelve was based on the stocks' prior month's average closing price in September 2020. The analysis for these twelve stocks used the same data source and methodology as the analysis in Tables 4 and 6, but was based on trading occurring between October 5-9, 2020. Because the new round lot size will be protected, this analysis also examines the change in the protected quotes under the final amendments.

1840 Commenters agreed that the new round lot definition would improve transparency. See, e.g., Schwab Letter at 4; CBOE Letter at 5. The Vendor Display Rule will require broker-dealers to show, in the context of which a trading or order-routing decision can be implemented, a consolidated display that includes quotes derived from the new round lot size. See supra Section II.D.2(b) and infra Section V.C.1(b)(vii).

market information, such as many retail investors. These market participants will benefit from being able to see information on these smaller quotes at better prices before they send in their orders, which may improve their trading decisions and order execution quality by providing an opportunity to realize gains from trade,[1841] as discussed below in this section.[1842] This may also improve price efficiency. This is because certain odd-lot information not currently disseminated as part of SIP data will be made available as part of core data; therefore market participants who use SIP data who previously did not use the information contained in these odd-lot quotes will be able to incorporate this information into their trading decisions. These trading decisions are integral to how market prices are formed. Also, the change may affect order routing and the share of order flow received by each exchange, since more market participants who rely on core data will be aware of quotes at better prices that are currently in odd-lot sizes, and these may not be on the same exchange as the one that has the best 100 share quote.

The Commission believes that changing the round lot definition to include smaller-size orders in stocks priced higher than $250 will benefit market participants who would have traded with price-improving odd-lot quotes in these stocks but do not do so because they cannot see information on odd-lot quotes.[1843] Under the final amendments, these market participants will be

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1841 See supra note 1593.

1842 It will also benefit market participants who post odd-lot quotes at prices superior to the NBBO because market participants that rely exclusively on SIP data may now be able to see some of these quotes and trade against them.

1843 Currently, some information about odd-lot quotes ends up in core data through certain exchanges rolling up odd-lot quotes into round lots. But even in this case, the rolled up quote is reported to the exclusive SIPs at the worst price out of all the odd-lots that were rolled up to produce the quote, so the full amount of price improvement available on that

able to see these quotes in core data, and make a decision about whether to trade based on this newly visible, improved price.[1844] This may benefit market participants, including many retail investors, because they will be able to realize the gains from trade that are available in this situation and are not currently occurring because of the lack of information. Also, some market participants may wish to exchange an odd-lot quantity of a stock by posting a limit order for an odd-lot amount. Currently, this order's price is not visible to market participants who rely solely on SIP data, and thus there may be delays in getting this limit order filled, since such market participants would not send market orders in. Thus, adding smaller-size quotes in core data for certain stocks will result in a benefit to both the market participants who would submit the market orders and the market participants who post the odd-lot quotes they execute against.

The magnitude of this benefit depends on the amount of additional trading generated by the inclusion of odd-lot information. In particular, the Commission believes that to the extent many market participants who rely solely on SIP data and lack information on odd-lot quotes would have traded frequently against odd-lot quotes had they known about them, the benefit will be large. However, if it is uncommon for market participants who would trade frequently against odd-lot quotes to rely solely on SIP data and to lack information on odd-lot quotes, then the Commission believes that the associated economic benefit from including smaller-size quotes in core data for certain stocks will be small. The Commission believes it is not possible to observe this willingness to trade with existing market data.

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exchange is still not visible to market participants relying solely on exclusive SIPs for market data.

[1844] Commenters agreed that the new round lot definition would show more odd-lot trading interest. See, e.g., CBOE Letter at 6; BlackRock Letter at 3.

The Commission believes that the new round-lot definition will benefit market participants by improving order routing in stocks priced higher than $250, provided that they do not already obtain information on odd-lots from proprietary feeds.[1845] Currently, market participants who rely on core data are not aware of odd-lot quotes available at other exchanges that exist at prices that are better than the national best bid and offer (e.g., the exchange with the best priced 100 share quote may not be the exchange with the best priced odd-lot quote).[1846] The new round lot definition will make more of these quotes in higher priced stocks visible to market participants that subscribe to core data, which will improve order routing and may improve order execution quality and facilitate best execution for these market participants.[1847]

The Commission believes that the new round lot definition may improve price efficiency for stocks priced above $250.[1848] The wider availability of information about odd-lot quotes may mean that more market participants (who currently rely solely on SIP data) will incorporate the

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1845 This benefit would apply to both market participants who are routing their own orders and market participants whose orders are being routed by a broker-dealer, provided the broker-dealer does not do not already obtain information on odd-lots from proprietary feeds.

1846 Battalio, Corwin, and Jennings (2016) examined the frequency of trading at inferior prices as compared to available unprotected odd-lot quotes in a sample of 10 high-priced stocks during one week in 2015. They found that there was an unprotected odd-lot limit order available at a better price for 2.52% of the trades that occurred. See Robert Battalio et al, *Unrecognized Odd Lot Liquidity Supply: A Hidden Trading Cost for High Priced Stocks*, 12 J. TRADING 35 (2016). A commenter also referenced this study and stated that unprotected odd-lot quotes at prices better than the NBBO at other exchanges get traded through. See BlackRock Letter at 4.

1847 For a discussion of order execution quality and the provision of execution services by broker-dealers, see supra Section V.B.3(e).

1848 For additional discussion of the price efficiency point, see infra Section V.D.1.

information contained in those quotes into their trading decisions. This may have the effect of improving the efficiency with which this information becomes reflected in prices.

The Commission believes that the new round lot definition may cause changes to order flow as market participants change their trading strategies to take advantage of newly visible quotes.[1849] This may mean that there would be changes to the share of order flow each exchange receives as a result of this rule. The Commission is uncertain about the magnitude of this effect.

As observed by commenters,[1850] the new round lot definition will also improve transaction cost analysis and best execution analysis in higher priced stocks, which are benchmarked against the NBBO. A smaller round lot size will improve these analyses because it will increase the accuracy of the NBBO, which will now better reflect smaller sized odd-lot quotes that may be available at better prices, possibly on another exchange.[1851]

Some commenters stated that new round lot tiers would increase complexity and create confusion among investors.[1852] The Commission acknowledges that the new round lot tiers may initially increase complexity when they are first implemented.[1853] However, after the new round

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1849 For example, currently a market participant, relying on SIP data, may submit an order to the exchange with the exclusive SIP NBBO and in the process, trade at an inferior price to an odd-lot quote that the market participant was not aware of on another exchange. If the market participant would have preferred to route to the price-improving odd-lot quote, then under the updates to core data the market participant will send the order to the exchange with the smaller, price improving quote.

1850 See, e.g., ICI Letter at 7; AHSAT Letter at 5. It will also improve the accuracy of Rule 605 statistics. See infra Section V.C.1(b)(vii).

1851 See supra Section V.C.1(b)(i) and discussion in this section.

1852 See, e.g., Clearpool Letter at 11–12; STANY Letter II at 3; TD Ameritrade Letter at 10.

1853 See infra Section V.C.1(b)(v).

lot tiers are implemented, the Commission does not believe they will significantly increase the complexity of the market or create confusion for a number of reasons.[1854] First, market participants already trade in stocks with round lot sizes other than 100 shares.[1855] Second, most NMS stocks will have a round lot size of 100 shares under the new round lot tier definitions.[1856] Third, core data will be distributed with the size of the NBBO and best quotes from each exchange given in shares and not number of round lots.[1857] Currently, the SIPs indicate size as the number of round lots available at the NBBO and each exchange's best quote, so investors need to convert round lot size to share size for stocks with round lots other than 100 shares. Under the final amendments, investors will observe the number of shares available and will not need to make this conversion. Fourth, the Commission expects that broker-dealers and other market participants will modify or develop their systems to automatically keep track of the different round-lot changes.[1858]

Commenters stated that the reduced round lot sizes would cause less liquidity to be available at the new NBBO in higher priced round lot tiers and that more marketable orders

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1854 Commenters agreed that the new round lot tiers would not add significant complexity. See, e.g., MEMX Letter at 4 ("Once tiers are required, although technology changes will be needed to implement the tiering structure, MEMX does not believe that there is significant additional complexity associated with supporting differing numbers of tiers."); IEX Letter at 4.

1855 See supra Section V.C.1(b)(i).

1856 The Commission estimates that approximately 98.5% of NMS stocks will have a round lot size of 100 shares. See supra Table 4.

1857 See supra Section II.C.2(e). One commenter stated that showing the number of shares rather than the number of round lots would reduce confusion with different size round lot tires. See CBOE Letter at 13–14.

1858 See infra Section V.C.1(b)(v).

would have to walk the book and execute at prices outside the NBBO.[1859] One of these commenters stated that the round lot tier structure in the Proposing Release would cause many retail investors' marketable orders to walk the book, which could lead to confusion and disappointment among retail investors because they are not used to having their orders walk the book.[1860] The Commission acknowledges that the smaller round lot size could lead to a smaller number of shares at the NBBO for most stocks in higher priced round lot tiers. However, this effect will depend on how market participants adjust their order submissions. For example, as observed by a commenter, orders pegged to the NBBO will remain at the NBBO.[1861] If this represents a significant portion of orders at the NBBO, then the number of shares at the NBBO may not change significantly.

If the size at the NBBO decreases in a stock in a higher priced round lot tier, then it could increase the frequency with which marketable orders walk the book. The adopted round lot tier sizes are based on a notional value of $10,000. Staff analysis estimated that the average notional trade size in 2019 was $8,068 (excluding auctions).[1862] Commenter analysis also observed that a

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1859 See, e.g., TD Ameritrade Letter at 7–8, 10; AHSAT Letter at 5; Nasdaq Letter III at 12.

1860 See, e.g., TD Ameritrade Letter at 8. The round lot tier sizes the commenter was referring to in the Proposing Release were based on a $1,000 notional size. The adopted round lot tiers are based on a larger $10,000 notional size, which should significantly decrease the frequently of a marketable order being larger than the notional value of the adopted round lot tiers compared to the round lot tiers in the Proposing Release. See supra Section II.D.2(a).

1861 See BestEx Research Letter at 6.

1862 See supra Section II.D.2(a). Several commenters stated that the Commission did not conduct analysis to determine the notional value of the proposed round lot tiers. See, e.g., Nasdaq Letter V at 4–5; Angel Letter at 13-14. In developing the notional value for the adopted round lot tiers, the Commission considered its estimate of the average trade

significant portion of trading occurred at or below $10,000.[1863] The Commission acknowledges that these estimates indicate that if the available liquidity at the NBBO is close to the $10,000 notional value, then there could be an increase in the frequency with which orders walk the book in the higher priced round lot tiers. However, as discussed above in this section, it is not entirely clear how investor orders and the size at the NBBO will change. Therefore, it is also uncertain how frequently orders will walk the book under the new round lot tiers. Even if the size at the NBBO declines, the Commission does not believe it will cause a significant increase in the frequency that retail investors' marketable orders walk the book and lead to confusion among retail investors for two reasons. First, currently most retail investor marketable orders execute off-exchange at retail internalizers and do not execute on an exchange and walk the limit order book.[1864] Because retail internalizers may offer price improvement, it is possible that the retail internalizer could fill the entire order at a price that is equal to or better than the NBBO.[1865] Second, even if a retail marketable order was routed to an exchange, it may not be greater than the notional size of the NBBO at an exchange in a higher priced round lot tier.[1866] Additionally,

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size in 2019 and commenter analysis on the size of trades and orders. See supra note 269 and accompanying text.

1863 See Virtu Letter at 3-4 (stating that data from 2019 to present show that the vast majority (over 75%) of all trades are still for less than $10,000); Angel Letter at 17 ("[T]he median trade size is roughly $10,000."); IntelligentCross Letter at 3 ("[T]he notional value of the median trade today is about $2,000.").

1864 See infra Section V.C.1(b)(iii).

1865 It is also possible that a retail internalizer could execute part of the order and route the rest to an exchange, where it could execute against the NBBO or walk the book.

1866 One commenter observed that the average retail trade size between 2007 and the present is around $14,581. See Virtu Letter at 3-4. The minimum notional size at the NBBO on a single exchange in the higher priced round lot tiers will be $10,000. If more than one

even if the size at the NBBO is smaller and a marketable order walks the book or a retail internalizer does not execute the whole order at the NBBO, the Commission does not believe that the average price at which it executes will decrease, i.e., transaction costs will not increase, because the NBBO will be at a better price.[1867]

As observed by commenters, protecting the smaller round lot quotes in higher priced stocks will benefit retail investors by better protecting their limit orders.[1868] One commenter observed that 23 percent of its customers' limit orders for stocks priced higher than $100 are less than 100 shares.[1869] Under the new round lot tiers, retail investors will benefit because a greater portion of their odd-lot sized orders in higher priced stocks will be protected and not traded-through.[1870]

One commenter stated that the smaller round lot size in higher priced stocks would disadvantage the limit orders of retail traders because it would make it easier for low-latency professional traders to step ahead of them with less risk.[1871] The Commission disagrees with this commenter. Currently low-latency professional traders that receive proprietary feeds that contain all odd-lot information do not need to submit a round lot sized order to step ahead of retail limit orders. These traders can submit an odd-lot-sized order to step ahead of the retail investor's order at a lower price and the retail investor may not observe it if the retail investor only receives SIP data. With the smaller round-lot size in higher priced stocks, retail investors

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exchange were at the NBBO, then an order would need to execute at the available liquidity at those exchanges before walking the book.

1867 See supra Section V.C.1(b)(i).

1868 See, e.g., Schwab Letter at 4–5; SIMFA Letter at 9–10.

1869 See Schwab Letter at 4–5.

1870 See supra note 1846.

1871 See TD Ameritrade Letter at 9.